to such depths. Or the molecules responsible for the stains might have been in the cometary fragments in the first place. We know from the *Vega 1* and *2* Soviet missions and the *Giotto* mission of the European Space Agency—both to Halley's Comet—that comets may be as much as a quarter composed of complex organic molecules. They are the reason that the nucleus of Halley's Comet is pitch black. If some of the cometary organics survived the impact events, they may have been responsible for the stain. Or, finally, the stain may be due to organic matter not delivered by the impacting cometary fragments, but synthesized by their shock waves from the atmosphere of Jupiter.

Impact of the fragments of Comet Shoemaker-Levy 9 with Jupiter was witnessed on seven continents. Even amateur astronomers with small telescopes could see the plumes and the subsequent discoloration of the Jovian clouds. Just as sporting events are covered at all angles by television cameras on the field and from a dirigible high overhead, six NASA spacecraft deployed throughout the Solar System, with different observational specialties, recorded this new wonder—the *Hubble Space Telescope*, the *International Ultraviolet Explorer*, and the *Extreme Ultraviolet Explorer* all in Earth orbit; *Ulysses*, taking time out from its investigation of the South Pole of the Sun; *Galileo*, on the way to its own rendezvous with Jupiter; and *Voyager 2*, far beyond Neptune on its way to the stars. As the data are accumulated and analyzed, our knowledge of comets, of Jupiter, and of the violent collisions of worlds should all be substantially improved.

For many scientists—but especially for Carolyn and Eugene Shoemaker and David Levy—there was something poignant about the cometary fragments, one after the other, making their death plunges into Jupiter. They had lived with this comet, in a manner of speaking, for 16 months, watched it split, the pieces, enshrouded by clouds of dust, playing hide-and-seek and spreading out in their orbits. In a limited way, each fragment had its own personality. Now they're all gone, ablated into molecules and atoms in the upper atmosphere of the Solar System's largest planet. In a way, we almost mourn them. But we're learning from their fiery deaths. It is perhaps